

BattleStats Test Plan

One battle, six level-18 characters, every kind of damage the stats page should be able to explain. Phase 1 only records; nothing shows in game.

Purpose

The recorder tags every damage, healing and block credit with how it arrived: a direct hit, a status tick (poison, bleed, burn), a tile's on-enter effect, a tile ticking at the start of a turn, a tile firing as it is placed, thorns, or a summon. At the end of the fight it prints each character's totals next to the numbers the game's own stats page uses. The test has one question to answer: does the game already count ticks and tiles for the caster, or does it drop them? Everything else (crits, kills, overkill, biggest hit, per-skill totals) is checked on the way.

Setup

1. **Host the game yourself** and bring all six characters in your own party. Recording happens on the machine that resolves the battle, so a solo party of six on one PC is the ideal test. Friends can join later runs, but for this one keep it local.
2. Respec each character into the build on its page (respec is 250 gold per level, 4,500 each at level 18; the purse has plenty). Point totals are exactly 20, the level-18 allowance, and tier unlocks are satisfied in order.
3. Equip what the build says. Casters' damage comes from the weapon's damage number, so give each caster the hardest-hitting staff or wand you own.
4. Config: `stolenrealm.battlestats.cfg` ships with `LogEveryHit = true` and `WriteJson = true`. Leave them. Verbose logging for the whole mod set is already on.
5. Pick a fight with **many enemies and at least one elite or champion**, at or below level 18, any difficulty. A big open map is better than a corridor: the tiles need enemies walking through them. Torturous is fine and gives the enemies enough health to sit in poison for a few turns.
6. Play the fight following the turn script (page 8). Take your time; there is no timing element.
7. When the post-battle screen appears the summary is written. Then send `BepInEx\LogOutput.log` and the newest file in `BepInEx\BattleStats\`.

What each character covers

Character	Build	Proves
1. Bloom	Nature: poison tiles, thorns, summons	Poison ticks credited to the caster; Bad Bloom tile; Brambles; Thorns; a summon's damage; poison spreading (Contagion); healing tile.
2. Ember	Fire: burning ground, burn ticks, fire thorns	Burning Ground turn-start ticks; Meteor's lingering fire; Rune of Exploding (tile placed); Immolate ticks; Fire Shield thorns; multi-target Fireball and Detonate.
3. Chaplain	Light: heals, shields, heal tile	HEAL lines; healing done vs received; self-heal from Kindred Spirit; Shield of Light blocks; Holy Ground heal tile; Shield of Retribution exploding.
4. Bonecaller	Shadow: summons, life steal, poison cloud, soul link	Three summons' damage routed to the master; life-steal healing; Poison Cloud tile; Soul Link duplicated damage; Haunt ticks.
5. Juggernaut	Warrior: two-hander, cleaves, bleed, self-damage	Multi-target hits; Bleeding ticks; Double Edge self-damage; Blood Drinker life steal; Colossus splash damage; being hit for the Thorns tests.
6. Cutthroat	Thief: crits, poison on hit, execute	CRIT lines; poison stacks from weapon hits; Deathblow kills and overkill; Smoke Bomb tile; max-damage Master Assassin.

1. Bloom

Nature · 20 points · the "my numbers look low" build

Gear: highest-damage staff or wand; any armor. **Stats:** Might and Vitality. Poison ticks scale from the weapon, not from Intelligence.

Tier	Skill	Why it is in the test	Pts
T1	Entangle	Direct physical hit plus 2 Poisoned stacks: a direct credit followed by ticks on later turns.	1
T1	Faerie Swarm	Physical damage every turn plus Bleeding: a second kind of status tick.	1
T1	Thorns I	Return Physical Damage: the "thorns" path when enemies hit Bloom.	1
T1	Brambles	Summoned barrier that hurts attackers: a summon and thorns at once.	1
T1	Beast Master I	Summon damage bonus (needed for T2 unlock anyway).	1
T1	Pack Hunter I	Filler damage bonus.	1
T2	The Bad Bloom	The key skill. A flower tile that applies 2 Poisoned stacks per turn to enemies in range. Every tick it causes must show as credited to Bloom.	1
T2	Venomous Skin	Cast on Juggernaut: attackers get poisoned. Tests whether poison applied through an ally's buff still credits Bloom.	1
T2	Thorns II	More thorns.	1
T2	Living Armor	Regeneration each turn: a healing tick (HEAL line every turn).	1
T3	Contagion	Poison spreads to neighbours: spread ticks must also credit Bloom.	1
T3	The Good Bloom	Healing tile: heals through a tile, the healing counterpart of Bad Bloom.	1
T3	Nature Summoning II	A real summon (stag, wolf or boar) whose hits should appear as Summon on Bloom.	1
T4	Mass Entangle	Multi-target direct hit plus 3 Poisoned stacks each.	2
T4	Circle of Life	Blooms spawned on enemy death: tiles created by a trigger rather than a cast.	2
T5	Titan Bloom	Buff tile; also confirms non-damage tiles produce no bogus HIT lines.	3

Expected in the log: HIT ... [StatusTick: Poisoned] and [StatusTick: Bleeding] credited to Bloom, [GroundEnter | The Bad Bloom] or [TurnStart | ...] when enemies stand in it, [Returned] when Bloom or the Brambles are hit, [Summon] for the animal, HEAL ... [StatusTick: Living Armor]. In the summary, Bloom's damage must equal the GAME'S OWN TOTALS DamageDealt. If it does not, the difference is what the game drops.

2. Ember

Fire · 20 points · burning ground and burn ticks

Gear: highest-damage staff or wand. **Stats:** Might and Vitality.

Tier	Skill	Why it is in the test	Pts
T1	Fireblast	Single-target direct fire hit: the baseline.	1
T1	Immolate	Fire damage each turn: a burn status tick.	1
T1	Fire Shield	Attackers take fire damage: fire thorns (Returned path with a fire type).	1
T1	Hot Head I	Damage bonus.	1
T1	Fire Borne	Filler.	1
T1	Burning Reach	Range, makes placing tiles easier.	1
T2	Burning Ground	The key skill. Enemies take fire damage each turn they stand in it. Ticks happen on Ember's turn start and must credit Ember.	1
T2	Fireball	Hits all targets in range: one HIT line per target expected.	1
T2	Hot Head II	Damage bonus.	1
T2	Fire Starter	Enemies start with Heat stacks; feeds Avatar of Flame.	1
T3	Aura of Flame	Continuous damage to enemies within 3 hexes: an aura tick, a path we have not classified yet. Note how it is labelled.	1
T3	Rune of Exploding	A tile that detonates a turn later: "tile placed" or "turn start", we want to see which.	1
T3	Detonate	Point-blank multi-target hit.	1
T4	Meteor	Direct blast plus lingering fire: one cast producing both a direct credit and tile ticks.	2
T4	Flash Fire	Shorter cooldowns so everything gets cast at least twice.	2
T5	Avatar of Flame	Each Heat stack deals fire damage per turn: a passive-driven tick with no cast behind it. Where does the game credit it?	3

Expected in the log: [TurnStart | Burning Ground] ticks at the start of Ember's turns, [StatusTick: Immolate], [Returned] when Ember is hit in melee, several HIT lines from one Fireball, Meteor as a direct hit followed by tile ticks. Watch for Aura of Flame and Avatar of Flame lines: their labels tell us whether the recorder needs a new path.

3. Chaplain

Light · 20 points · healing, shields, heal tile

Gear: highest-damage staff or wand (heals scale from it too), a shield if you have a one-hander. **Stats:** Vitality and Might.

Tier	Skill	Why it is in the test	Pts
T1	Cure	Plain single-target heal: HEAL line, HealingAdministered on Chaplain, HealingReceived on the target.	1
T1	Regenerate	Heal per turn: a healing tick.	1
T1	Healing Hand	Heals 30% of max health: a percentage heal.	1
T1	Blessed By Light I	Mana.	1
T1	Empowered Light I	Healing bonus.	1
T1	Elemental Protection	Buff; confirms buffs create no credit lines.	1
T2	Shield of Light	Key. Absorbs damage: BLOCK lines of type BlockedByShields on the shielded ally.	1
T2	Bless	Party buff.	1
T2	Kindred Spirit	Healing others also heals you 20%: a self-heal credit generated by a heal.	1
T2	Seal of Protection	Damage reduction: BlockedByDamageReduction lines.	1
T3	Mass Cure	Multi-target heal: one HEAL per ally.	1
T3	Light's Strength	Might bonus.	1
T3	Light's Beckon	Utility: pull an enemy into Burning Ground or the Bad Bloom.	1
T4	Holy Ground	Key. Heal tile: healing through a tile per turn, the mirror image of the damage tiles.	2
T4	Shield of Retribution	Absorbs, then explodes for fire damage when it breaks: damage credited from a broken shield. Put it on Juggernaut and let it break.	2
T5	Salvation	Extra status on heal targets; confirms status application alone produces no credit.	3

Expected in the log: HEAL ... HealingAdministered for Chaplain and matching HealingReceived on the target; BLOCK ... BlockedByShields on whoever carries Shield of Light; Holy Ground heals labelled as a tile; a fire HIT credited to Chaplain when Shield of Retribution pops. In the summary, healing done should equal the sum of the targets' healing received (plus Kindred Spirit self-heals).

4. Bonecaller

Shadow · 20 points · summons, life steal, poison cloud, soul link

Gear: highest-damage staff or wand. **Stats:** Might and Vitality (Soul Crush also scales with max health).

Tier	Skill	Why it is in the test	Pts
T1	Raise Skeletal Archer	Key. A ranged summon: its hits should appear as Summon on Bonecaller and in the game's SummonDamageDealt.	1
T1	Necromancer I	Summon bonus.	1
T1	Tainted Touch	Shadow hit with 12% life steal: a direct hit that also produces a HEAL on the caster.	1
T1	Call of the Grave	Damage bonus below 50%: finishing hits.	1
T1	Ghost Armor	Negates the next attack: does a fully negated hit produce a BLOCK line or nothing?	1
T1	Blind	Filler for the unlock.	1
T2	Poison Cloud	Key. A tile that applies 4 Poisoned stacks within 2 hexes: a second poison-tile source, from a different tree than Bloom's.	1
T2	Necromancer II	Summon bonus.	1
T2	Thirst I	Life steal on everything: HEAL lines paired with hits.	1
T2	Haunt	Shadow damage per turn with life steal: a tick that also heals.	1
T3	Raise Skeletal Warrior	Melee summon; try to let it land a killing blow (KILL line should name Bonecaller or the skeleton, we want to see which).	1
T3	Soul Crush	Big single hit for the biggest-hit stat.	1
T3	Consumption	Multi-target drain.	1
T4	Vampiric Aura	Allies life steal: healing on other characters caused by Bonecaller's aura.	2
T4	Soul Link	Key. Damage to one target is also dealt to the linked one: duplicated credits, who gets them?	2
T5	Raise Skeletal Mage	Third summon, caster type.	3

Expected in the log: HIT ... [Summon] lines on Bonecaller for every skeleton hit and a matching SummonDamageDealt in the game totals; HEAL after life-steal hits; [StatusTick: Poisoned] from Poison Cloud credited to Bonecaller; with Soul Link active, two credits per hit on the linked pair.

5. Juggernaut

Warrior · 20 points · two-hander, cleaves, bleed, self-damage, the party's punching bag

Gear: the biggest two-handed weapon you have, heavy armor. **Stats:** Might and Vitality. Stand in front so enemies hit you: that is what fires Thorns, Venomous Skin, Fire Shield and the shields.

Tier	Skill	Why it is in the test	Pts
T1	Double Edge	Key. 1.8x weapon damage and 15% back to you: a hit and a self-inflicted TAKEN line in one cast.	1
T1	Bash	Knockback: push an enemy into Burning Ground or the Bad Bloom (tile on-enter credit for Ember or Bloom).	1
T1	Fracture	Direct hit that lowers resistance: BLOCK-by-resistance lines should shrink afterwards.	1
T1	Rage	Damage up, damage taken up: more hits on the tank.	1
T1	Warmonger I	Damage bonus.	1
T1	Challenger	Armor when surrounded: BlockedByArmor lines.	1
T2	Cleave	Arc hit on several enemies.	1
T2	Warmonger II	Damage bonus.	1
T2	Tempered Rage	Rage variant with armor.	1
T2	Into the Fray	Damage per nearby enemy.	1
T3	Bleeding Cleave	Key. Arc hit applying 2 Bleeding stacks to each target: a burst of direct credits then bleed ticks on several enemies at once.	1
T3	Slam	Line hit, another multi-target shape.	1
T3	Two-Handed Mastery	Damage bonus.	1
T4	Bone Collector	Stacks on kills: make sure Juggernaut gets a few kills.	2
T4	Blood Drinker	Basic attacks life steal: HEAL on basic hits.	2
T5	Colossus	Key. Damage splashes to neighbours for 50%: the game records this splash specially; we want to see how it is labelled and to whom.	3

Expected in the log: a HIT and a TAKEN on Juggernaut for each Double Edge; several HIT lines per Cleave and Slam; [StatusTick: Bleeding] ticks; extra splash HIT lines on neighbours with Colossus; BLOCK ... BlockedByArmor when enemies hit him; [Returned] credits going to Bloom (Venomous Skin poison) when he is the one being hit, not to him.

6. Cutthroat

Thief · 20 points · crits, poison on hit, execution

Gear: two daggers or a dagger and a one-hander, whatever gives the highest crit chance. **Stats:** Dexterity heavy (crit chance and crit damage), rest Vitality.

Tier	Skill	Why it is in the test	Pts
T1	Poison Weapon	Key. Every attack applies Poisoned: poison ticks that originate from a weapon buff rather than a spell.	1
T1	Death Dealer I	Crit damage.	1
T1	Cripple	Direct hit with a status.	1
T1	Ambush I	Damage bonus on isolated enemies.	1
T1	Fencer's Finesse I	Crit chance.	1
T1	Dagger Throw	Ranged direct hit.	1
T2	Death Dealer II	Crit damage.	1
T2	Poisoned Dagger	Hit plus 2 Poisoned stacks.	1
T2	Ambush II	Damage bonus.	1
T2	Hide In Shadows	Stealth: does the first hit out of stealth get flagged differently?	1
T3	Garrote	Weapon hit plus 3 Bleeding stacks and Disabled.	1
T3	Hidden Blade	Basic-attack crit damage: big CRIT lines.	1
T3	Fight Dirty	2 more Poisoned stacks per attack.	1
T4	Smoke Bomb	A non-damage tile: must produce no HIT lines at all.	2
T4	Master Assassin	Always max damage: hit amounts should be identical for repeated casts on the same target, easy to spot.	2
T5	Deathblow	Key. Crits deal +200%: use on a low enemy for a KILL with a large overkill value.	3

Expected in the log: CRIT lines under a good share of Cutthroat's hits; [StatusTick: Poisoned] ticks credited to Cutthroat on every enemy he has touched; a **KILL ... by Cutthroat (Deathblow, Direct)** with **overkill**; no lines from Smoke Bomb. In the summary, crit rate should match what you saw on screen.

The battle, turn by turn

Order within a turn does not matter. The goal is that every path fires at least twice and that a few enemies stay alive long enough to take several ticks.

Before the first move

- Note which enemies are where. Pick two enemies that will be kept alive as "tick sponges": poisoned, burning, bleeding and standing in tiles. Do not focus them.
- Juggernaut takes the front; everyone else stays behind Brambles.

Turn 1: lay the ground

- **Ember:** Burning Ground across the lane enemies will walk through.
- **Bloom:** The Bad Bloom next to the largest group. Brambles between the group and the party.
- **Chaplain:** Holy Ground under the party. Shield of Light on Juggernaut.
- **Bonecaller:** Raise Skeletal Archer. Poison Cloud on the largest group.
- **Juggernaut:** Rage, then Double Edge on the nearest enemy (watch for the self TAKEN line).
- **Cutthroat:** Poison Weapon, then Dagger Throw at a tick sponge.

Turn 2: statuses and multi-target

- **Ember:** Immolate a tick sponge. Fireball the group. Rune of Exploding under a cluster.
- **Bloom:** Entangle the other sponge. Venomous Skin on Juggernaut. Nature Summoning II.
- **Chaplain:** Regenerate on Juggernaut. Cure whoever took damage. Light's Beckon an enemy into Burning Ground.
- **Bonecaller:** Haunt a sponge. Raise Skeletal Warrior. Tainted Touch anything.
- **Juggernaut:** Bleeding Cleave into the group. Bash an enemy into a tile.
- **Cutthroat:** Garrote a sponge. Basic attack something else (poison from the weapon).

Turn 3: the big shapes

- **Ember:** Meteor on the group (direct hit plus lingering fire). Detonate if surrounded.
- **Bloom:** Mass Entangle. The Good Bloom under the party.
- **Chaplain:** Mass Cure. Shield of Retribution on Juggernaut, then let it break.
- **Bonecaller:** Soul Link the two sponges. Consumption. Raise Skeletal Mage.
- **Juggernaut:** Slam through a line. Basic attacks (Blood Drinker life steal).
- **Cutthroat:** Smoke Bomb on the party. Deathblow on the lowest enemy: aim for a kill with overkill.

Turn 4 onward: let it run

- Keep the sponges alive and inside the tiles; re-apply Immolate, Haunt, Entangle as they expire. Let the skeletons and the animal fight; try to let a summon take a kill.
- Take a few hits on Juggernaut and on Bloom so Thorns, Fire Shield and Venomous Skin all fire.
- Finish with Cutthroat and Juggernaut taking kills so both have kills in the summary. Then let the sponges die to ticks if you can: a **KILL ... (Poisoned, StatusTick)** line is the best possible evidence.

When the post-battle screen appears, open the stats tab and glance at it, then quit to the menu. The summary and the JSON are written the moment that screen opens.

Checklist and what to send

✓	Check
	Log has ===== BATTLE START (turn 1) ===== when the fight begins.
	Every floating damage number has a HIT line; crits have a CRIT line under them; Fireball, Cleave, Slam, Mass Cure give one line per target.
	Poison, bleed, burn and Haunt ticks appear as [StatusTick: <name>] credited to the caster, not to the enemy and not missing.
	Burning Ground shows [TurnStart Burning Ground]; Bad Bloom and Poison Cloud show as a tile or as status ticks credited to their owner.
	Thorns, Fire Shield and Venomous Skin produce [Returned] credits for the character who was hit (Bloom, Ember) and poison ticks for Bloom.
	Skeleton and animal hits show as [Summon] on their master.
	Every enemy death has a KILL <enemy> by <name> line naming the right character and path; Deathblow shows overkill.
	Heals give HEAL lines; Shield of Light gives BLOCK ... BlockedByShields; Ghost Armor either gives a BLOCK line or nothing (note which).
	The summary's turn count equals the last turn number you saw.
	The decisive check: in the summary, for each character, damage equals the GAME'S OWN TOTALS DamageDealt (plus SummonDamageDealt and DamageReturned where present). Write down any character where they differ and by how much.
	No record failed warnings anywhere in the log.

Things worth writing down while you play

- Any damage number you saw on screen that has no matching HIT line (who, roughly how much, what caused it).
- Any line whose path label looks wrong to you: for example a Burning Ground tick labelled Direct, or a poison tick labelled with the wrong caster.
- How Aura of Flame, Avatar of Flame, Colossus splash and Soul Link were labelled: those are the four we could not predict from the code.

Send me

- BepInEx\LogOutput.log (the whole file; it keeps sessions, the last SESSION START is yours).
- The newest file in BepInEx\BattleStats\ (one JSON per battle).
- Your notes from above, and a screenshot of the game's stats page if you can.

Respec cost is 250 gold per level; all six builds together are about 27,000 gold. Skill totals are 20 points each, which is the level-18 allowance (level minus 1, plus 3). Tier unlocks need 2, 4, 6 and 10 points in earlier tiers; every build spends 6 in tier 1 and 4 in tier 2 before going higher, so nothing is locked.